MECH MAINT Department
SIMHADRI

**Ref:**SMTPP/MECH MAINT/2022-23/LMI/33587 **Date:**13/04/2022(16:22:14)

**Subject:**Approval for Revision of LMI



The Location Management Instruction (LMI) of Thermal Performance of Cooling Towers i.e. LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006 revision is done according to the external LMI audit exception points.

In this LMI the calculation sheet has been changed.

The approving authority for LMI is HOP.

Submitted for kind approval.

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कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

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**NTPC Ltd**

**LOCATION MANAGEMENT INSTRUCTION**

# *Thermal Performance of Cooling Towers*

**LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006**
**Rev:06 APR:2022**

Approved for Implementation by: **HOP** (**SMSTPS)**

Enquiries to: **HOD (EEMG)**

| | |
|---|---|
| **TITLE** | **Thermal Performance of Cooling Towers at NTPC Simhadri** |
| **REF NO** | **LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006 Rev:06 Apr 2022** |

# LMI APPROVAL DOCUMENT

| | | |
|---|---|---|
| 1 | LMI NAME | Thermal Performance of Cooling Towers at NTPC Simhadri |
| 2 | LMI No. | LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006 Rev: 06 |
| 3 | Reference Document | OGN/OPS/PERF/006/ Rev:06 Apr:2022 |
| 4 | Superseded Document | LMI/SMSTPS/OGN/OPS/PERF/006/ Rev:05 Apr:2021 |
| 5 | Prepared By | Name: Puspita Behera<br>Designation: Sr. Manager (EEMG) |
| 6 | Checked By | Name: Jhimli Dhar<br>Designation: AGM (EEMG) |
| 7 | Reviewed By | Name: I Laxma Reddy<br>Designation: AGM (MTP & EEMG) |
| 8 | Approved By | Name: Girish Chandra Choukse<br>Designation: GGM (Simhadri) |

| TITLE | Thermal Performance of Cooling Towers at NTPC Simhadri |
|---|---|
| REF NO | LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006 Rev:06 Apr 2022 |

# INDEX


| SL NO | CONTENT | PAGE NO |
|---|---|---|
| 1 | Introduction | 4 |
| 2 | Objective | 4 |
| 3 | Superseded Document | 4 |
| 4 | Records of Revision | 4 |
| 5 | CW system in Simhadri | 4 |
| 6 | Role of Cooling Towers in power station | 4 |
| 7 | Salient terms used in CT testing | 5 |
| 8 | Test Procedure | 5 |
| 9 | Performance of Cooling Tower | 7 |
| 10 | Evaluation of Cooling Tower Performance | 8 |
| 11 | Frequency | 9 |
| 12 | Cooling Tower Report Format | 9 |
| 13 | Review | 10 |

| TITLE | Thermal Performance of Cooling Towers at NTPC Simhadri |
|---|---|
| REF NO | LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006 Rev:06 Apr 2022 |

## 1.0 INTRODUCTION

Cooling water system plays a vital role in dissipation of waste heat in power station. More than 60 % of total heat input to the plant is finally dissipated as waste heat. The waste heat from the power plant is carries away by circulating water and ultimately gets dissipated in cooling tower.

## 2.0 OBJECTIVE

The purpose of this test is to describe procedure for testing and performance evaluation of Cooling Tower.

## 3.0 SUPERSEDED DOCUMENTS

This supersedes LMI/OGN/OPS/PERF/006: Thermal Performance of Cooling Towers at NTPC, in Revision 5 issued in Apr: 2021.

## 4.0 RECORDS OF REVISION

Entire LMI has been revised according to the external Audit exception points.

## 5.0 CW SYSTEM

CW system at NTPC Simhadri is closed recirculating type and having Natural draught Cooling Towers (NDCT).

Four Natural Draft cooling towers have been installed for 4 x 500 MW units of NTPC Simhadri which are of counter-flow type with Splash fills.

## 6.0 ROLE OF COOLING TOWER IN POWER STATIONS

Cooling towers play an important role in ensuring the thermal efficiency of a generating station. Thermal efficiency of a steam turbine is dependent on the temperature of the exhaust steam from the LP turbine into the condenser. Heat from the steam side of the condenser is transferred to the circulating water. This heat is in turn transferred to the atmosphere in cooling tower.

In case the cooling tower does not perform satisfactorily, the cold-water temperature of the cooling water would go up. This in turn would affect the condensing temperature of steam and back- pressure in the condenser. This would result in an increased heat rate and subsequent increase in fuel input.

| TITLE | Thermal Performance of Cooling Towers at NTPC Simhadri |
|---|---|
| REF NO | LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006 Rev:06 Apr 2022 |

## 7.0 SALIENT TERMS USED IN CT TESTING:

### 7.1 Approach:

Difference between the Cold-Water Temperature at CT outlet and inlet air Wet-Bulb Temperature.

### 7.2 Range:

Difference between the Hot Water Temperature (inlet to CT) and Cold-Water Temperature (outlet of CT)

### 7.3 Tower Capability:

The most reliable means to assess the cooling tower thermal performance is tower capability. It is defined as the percentage of water that the tower can cool to the design cold water temperature when the inlet wet-bulb, cooling range, water flow rate is all at their design value.

### 7.4 Effectiveness:

$$\text{Effectiveness} = \frac{\text{Range} \ * 100}{\text{Range + Approach}}$$

Range = Hot water Temp - Cold water Temp

Approach = Cold water Temp - Wet bulb Temp

## 8.0 TEST PROCEDURE:

### 8.1 Parameters to be measured:

**a) Circulating Water Flow:**

The water flow can be measured by measuring the CW pump head and evaluating the flow from the pump characteristic curves or by pitot tube. The flow measurement can be done in the pipe at pitot tube pit which is sufficient for traversing up to the center of 3.2-meter header. Refer to PG test procedures for method of calculations.

**b) Hot Water Temperature measurement:**

The thermometers may be inserted into the channel where water distribution network is open or into the thermo wells provided on the inlet riser.

| TITLE | Thermal Performance of Cooling Towers at NTPC Simhadri |
|---|---|
| REF NO | LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006 Rev:06 Apr 2022 |

**c) Cold Water Temperature:**

The re-cooled water temperature can be measured directly at the point where the circulating water is discharged from the basin, the average cold-water temperature being determined by simultaneous test readings taken across the selected sections. The measurement location is selected such that proper mixing of the cold water is ensured. For getting representative measurement the cold-water temperature, grid arrangement (see Fig-1) may be made at measuring locations across the cold-water channel. RTD's may be installed at 4-5 locations, across the horizontal plane and temperature readings taken using digital temperature read out connected to RTD'S. Grid arrangement for Cold Water Temperature Measurement.

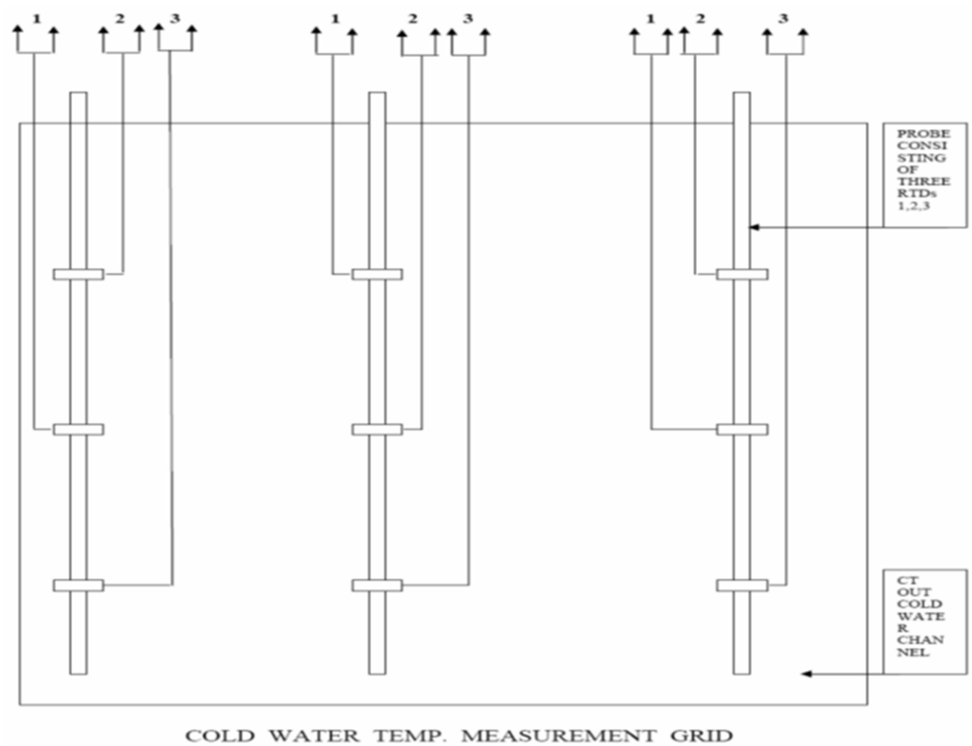


FIG-1

**d) Ambient Wet Bulb Temperature:**

Inlet wet bulb temperature is measured with mechanically aspirated psychrometer having mercury in glass thermometer with 0.1ºC graduation at a distance of 100 meters from the equipment in the direction of wind at height of 1.5 meters from ground equally spread along a line breaking the flow of air to the tower not directly under the sun and the wet bulb wick kept dipped in water.

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|---|---|
| **TITLE** | **Thermal Performance of Cooling Towers at NTPC Simhadri** |
| **REF NO** | **LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006 Rev:06 Apr 2022** |

**e) Dry Bulb Temperature**:

The dry bulb temperature is measured with mechanically aspirated psychrometer having 0.1°C graduated mercury in glass thermometer. The location is same as above.

### 8.2 Test Conditions:

The following variations from design condition shall not be exceeded,

| | | |
|---|---|---|
| 1 | CW flow rate | 90-110% of Design |
| 2 | Cooling Range | 80-120% of Design |
| 3 | Wet-bulb Temp | Design +/-8.5 C |
| 4 | Wind Velocity | Average 4.5 m/s<br>One-minute Duration:7.0 m/s |
| 5 | Dry bulb Temp | Design +/- 10 Deg C |
| 6 | Relative Humidity | Shall not fall below 40% |

a) Hot water distribution should be uniform throughout the tower.
b) Water level in cold water basin shall be at normal operating level and shall be maintained substantially constant during the test.
c) At the time of conducting the test the tower shall be in good operatingcondition.

### 8.3 Constancy of Test Conditions:

Variation In test condition shall be within the following limits

Circulating water flow shall not vary by more than 2 %
Heat load should not vary more than 5 %

### 8.4 Duration and frequency of measurements:

Hot water temperature, cold water temperature, dry bulb and wet bulb temperature for humidity shall be measured at 5 minutes intervals for a period of one hour.

## 9.0 PERFORMANCE OF COOLING TOWER

### 9.1 Parameters Affecting the Performance

The following parameters affect the performance of cooling tower:

(i) Wet bulb temperature

(i) Relative humidity or dry bulb temperature

(ii) Water flow rate

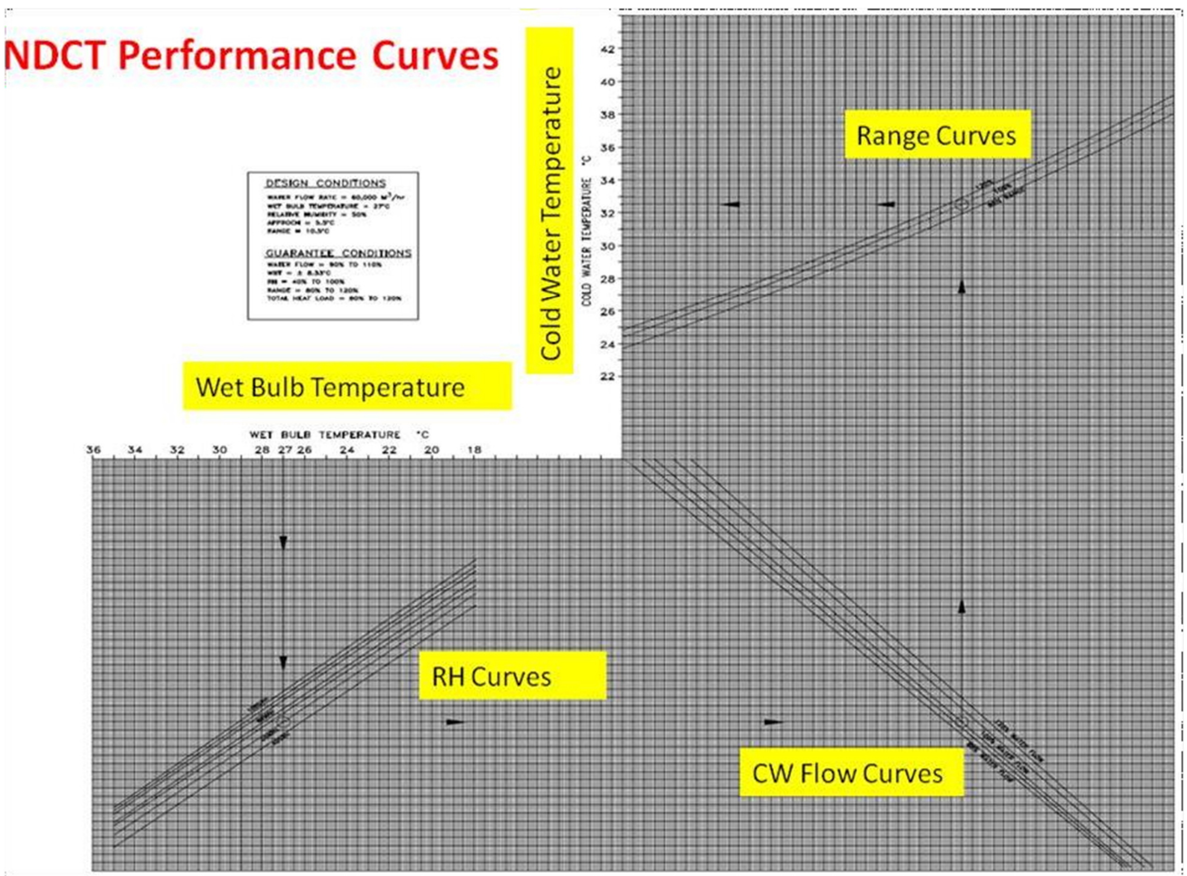
| TITLE | Thermal Performance of Cooling Towers at NTPC Simhadri |
|---|---|
| REF NO | LMI/SMSTPS/EEMG-08/OGN/OPS/PERF/006 Rev:06 Apr 2022 |

## 10.0 EVALUATION OF COOLING TOWER PERFORMANCE:

### 10.1.1 Plant Specific data and Characteristics Curves

Reference curves provided by manufacturer (Wet bulb temperature and cold-water temperature) for difference design ranges and at 90%, 100% and 110% of circulating water flow used for evaluation of tower capability.

### 10.1.2 Steps for determining Predicted Cold-water Temperature& Predicted CW Flow Rate (NDCT)




The above curve represents the thermal behavior of the cooling towers as a function of the ambient air state and the hydraulic and thermal loads.

a) Wet bulb temperature vs relative humidity gives a value Y in curve at point of intersection.
b) Line drawn parallel to x-axis from value Y vs water flow inclined lines gives a value X at point of intersection.

c) Line drawn parallel to y-axis from value X vs Range inclined lines gives cold water temperature at point of intersection.

Range =measure HWT-measure CWT

For a given wet bulb temperature (say 27.8 °c ) & RH (say 60% RH inclined lines ) a line is drawn from point of intersection parallel to x-axis to water flow lines (say 100% water flow inclined lines) a line is drawn from point of intersection parallel to y-axis to range lines(Say 11 °c inclined lines) , a line is drawn from point of intersection parallel to x-axis to get Predicted cold water temperature (32.5 °c).

### 10.1.3 Thermal Performance

a) Take readings of inlet Dry/Wet bulb temperature, Hot/Cold water temperatures and cooling water flow. Then check with the cooling tower performance curves for predicted cold water temperature. The difference between predicted cold-water temperature and actual cold-water temperature must be compared with design value to evaluate the performance of the tower.

Responsibility: HOD (EEMG)

b) Performance improvement factor:

It is the deviation between Predicted cold-water temperature and actual cold-water temperature during the test. If it is </= 0.3 °c, the tower performance said to be satisfactory.

Responsibility: HOD (EEMG)

## 11.0 FREQUENCY

Once in a year and pre/post overhaul of the unit

Responsibility: HOD (EEMG)

## 12.0 Cooling Tower Report Format:

| **Parameters** | **Design** | **Test** |
|---|---|---|
| Water flow (m3/hr) | | |
| Hot water temperature (Deg C) | | |
| Cold water temperature (Deg C) | | |
| Wet bulb temperature (Deg C) | | |
| Dry bulb temperature (Deg C) | | |
| Range (Deg C) | | |

| | | |
|---|---|---|
| Approach (Deg C) | | |
| Effectiveness (%) | | |
| Predicted Cold water temperature (Deg C) | | |
| Shortfall in Cold water temperature (Deg C) | | |

## 13.0 REVIEW

This Guidance Note shall be reviewed by HOP (Simhadri) every 2yrs or as andwhen necessary.